

Q1 (2022 T1 Midterm Exam Q3)

The circuit in Figure 1 corresponds to a bicycle with front and rear bulb lights and a permanent magnet generator (hub dynamo) to power them. The circuit also includes a USB charger to charge the mobile phone while riding.

When the bicycle is travelling at 20 km/h (average cycling speed), the permanent magnet generator voltage is 5 V.

- Use superposition to calculate the current flowing through the front bulb i_{front} .
- Calculate the power dissipated by the front bulb (i.e., $20\ \Omega$ resistor).

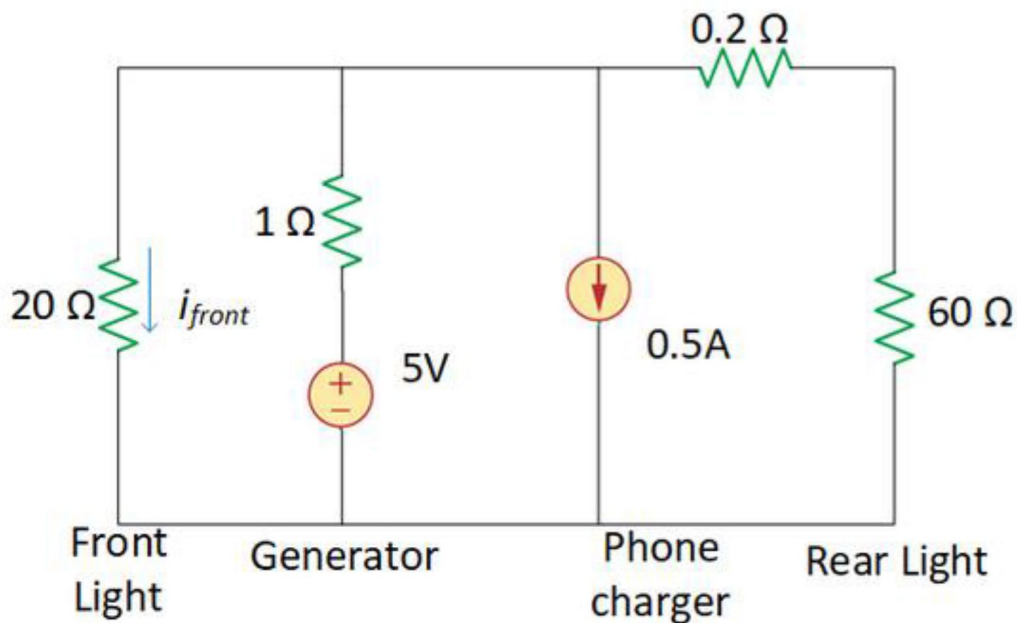


Figure 1

Answers: i) $i_{front} = 0.21\text{ A}$, ii) 0.882 W

Q2 (2023 T3 Midterm Exam Q4a)

For the circuit shown in Figure 2, use a succession of source transformations (only source transformations and resistor/source simplification) to find an equivalent circuit consisting of a single current source and a single resistor as seen from terminals a-b. Then, determine the Norton current (I_N) and Norton resistance (R_N) at the terminals a-b.

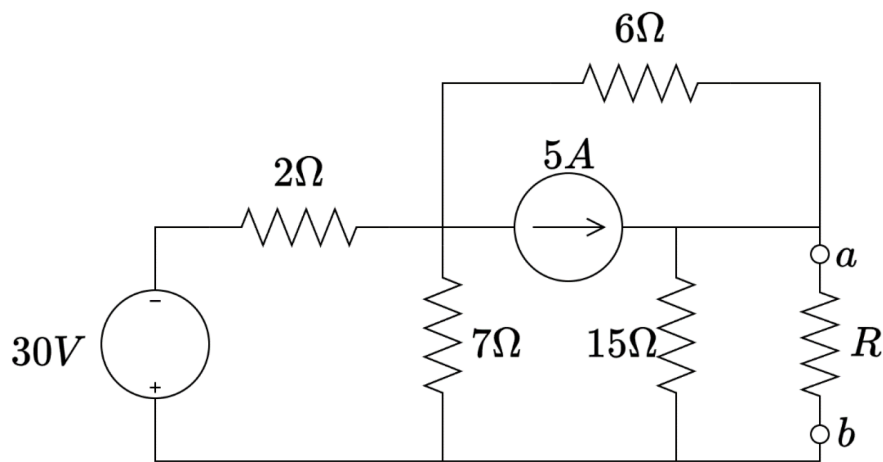


Figure 2

Answers: $I_N = -0.88\text{A}$, $R_N = 5\ \Omega$

Q3 (2018 T3 Midterm Exam Q4)

The system shown in Figure 3 is being used to power a load.

- Find the Thevenin equivalent of the system.
- Find the power in the load using the Thevenin equivalent model from part i).

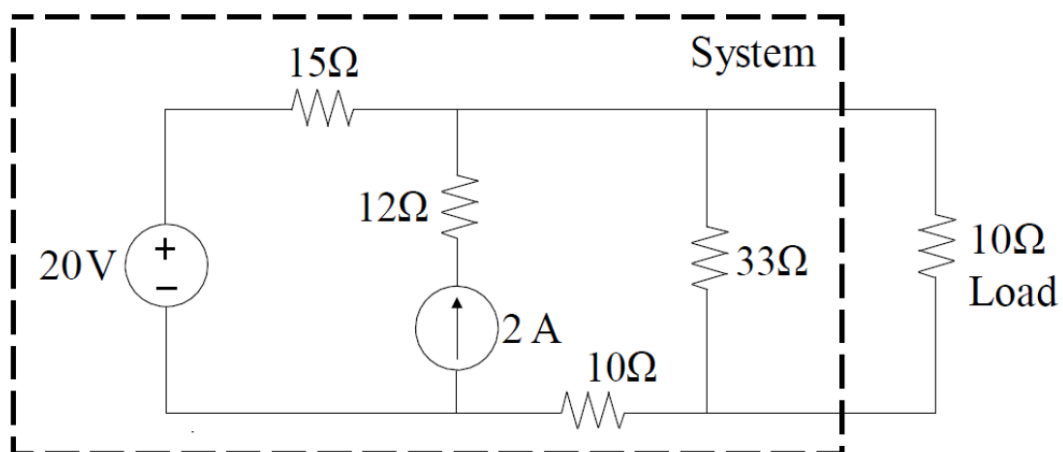


Figure 3

Answers:

- i) $V_{th} = 28.45\text{ V}$, $R_{th} = 14.22\ \Omega$, ii) $P_{load} = 13.8\text{ W}$

Q4 (Modified from 2023 T3 Midterm Exam Q4b)

The circuit shown in Figure 4 is being used to power a ventilation system. The ventilation fan can be modelled as a single resistor connected to terminals a-b, as per the figure.

- Calculate the Thevenin voltage at terminals a-b.
- Calculate the value of the fan resistance that will ensure the maximum transfer of power from the circuit to the fan.
- Find the maximum power that can be delivered to the fan.
- Draw the Norton equivalent circuit at the terminals a-b.
- Recalculate the resistance of the fan necessary for maximum power transfer using another valid method.

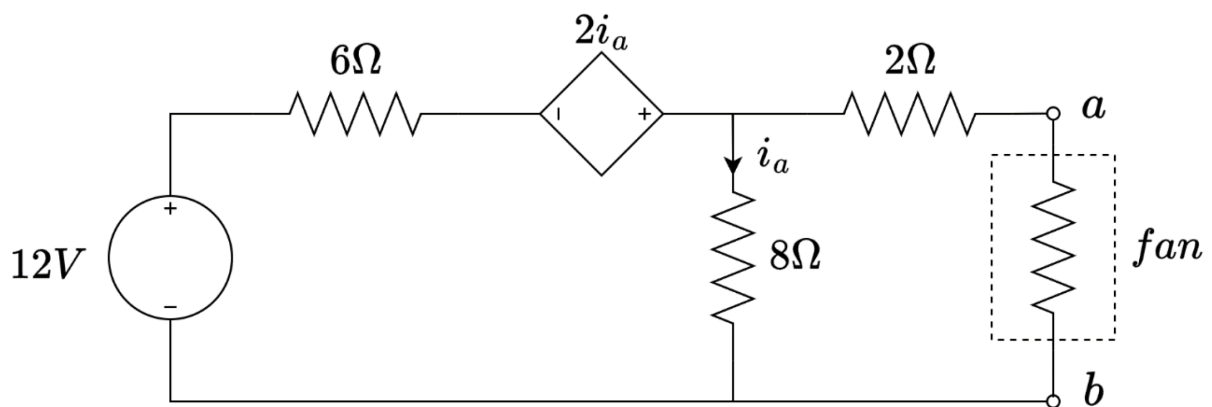


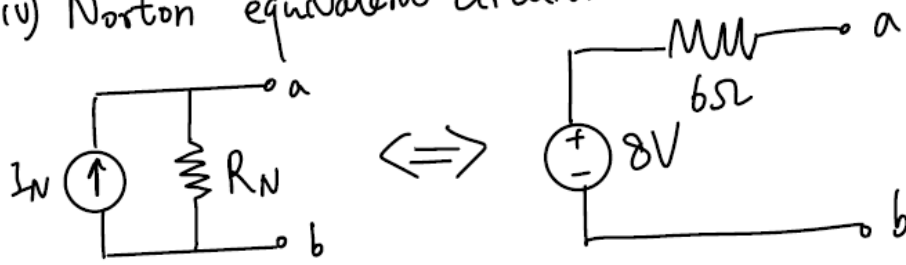
Figure 4

Answers:

- i) $V_{th} = 8 \text{ V}$, ii) $R_{th} = 6 \text{ } \Omega$, iii) $P_{max} = 2.667 \text{ W}$, iv) $I_N = 1.333 \text{ A}$, $R_N = 6 \text{ } \Omega$, v) $R_{th} = 6 \text{ } \Omega$

Solutions to the modified questions:

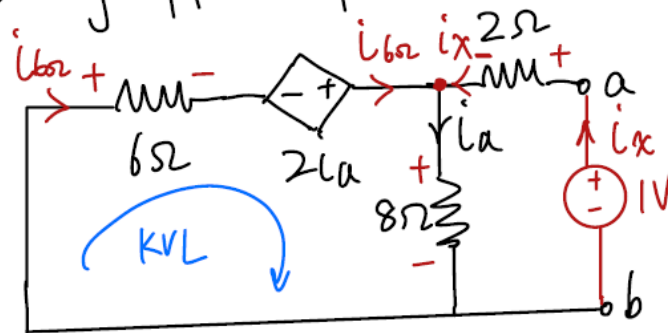
iv) Norton equivalent circuit:



$$\begin{cases} I_N = \frac{8}{6} = 1.33 \text{ A} \\ R_N = 6 \Omega \end{cases} \quad (\text{Source transformation})$$

The solution provided in the past exam paper used Method 2. Therefore, in this instance, we will demonstrate how to calculate R_{th} using Method 3.

Turning off independent sources:



$$R_{th} = \frac{1}{i_x}$$

KVL @ left mesh:

$$+6i_{6\Omega} - 2i_a + 8i_a = 0 \quad (1)$$

KVL @ right mesh:

$$-8i_a - 2i_x + 1 = 0 \quad (2)$$

KCL

$$i_{6\Omega} + i_x = i_a \quad (3)$$

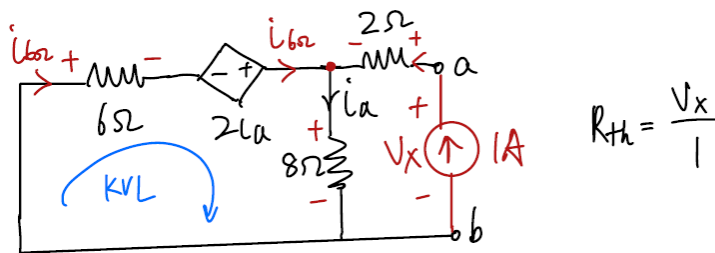
$$\textcircled{3} \rightarrow \textcircled{1}: 6(i_a - i_x) - 2i_a + 8i_a = 0$$

$$\begin{cases} 12i_a - 6i_x = 0 & \textcircled{4} \\ 8i_a + 2i_x = 1 & \textcircled{5} \end{cases}$$

Solving $\textcircled{4}$ & $\textcircled{5}$: $\begin{cases} i_a = \frac{1}{12} \text{ A} \\ i_x = \frac{1}{6} \text{ A} \end{cases}$

$$R_{th} = \frac{1}{i_x} = 6 \Omega$$

You can also use Method 3 by attaching a current source (e.g., 1 A):



$$\text{KCL: } i_{6\Omega} + 1 = i_a \quad \textcircled{1}$$

KVL @ Left mesh:

$$6i_{6\Omega} - 2i_a + 8i_a = 0 \quad \textcircled{2}$$

Substitute $\textcircled{1} \rightarrow \textcircled{2}$:

$$6(i_a - 1) - 2i_a + 8i_a = 0$$

$$i_a = \frac{6}{12} = 0.5 \text{ A}$$

KVL @ right mesh:

$$-8i_a - 2 \times 1 + V_x = 0$$

$$V_x = 8i_a + 2 = 4 + 2 = 6 \text{ V}$$

$$R_{th} = \frac{V_x}{1} = 6 \Omega$$